Module-5

Multi-Level Watermark for IP Protection

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Introduction

- An Integrated Circuit (IC) is designed using various abstraction levels of design flow.
- The multi-level watermark is deployed using multiple abstraction levels.
- In the VLSI design flow of application specific integrated circuits (ASICs), the system specifications represent the design at the highest level of abstraction.
- The next level of the design flow is the architectural or behavioural level, also referred as electronic system level (ESL).
- The design process at this level is integrated with the 'design space exploration (DSE)' technique to identify an optimal resource configuration.
- A high-level design process results into the register transfer level (RTL) description of the design flow.
- The subsequent levels of the design flow are the logic level and physical level.

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- In the context of complex application such as DSP cores, ESL and RTL may be more significant levels of the design flow during optimization of design objectives.
- Moreover, these levels offer high flexibility than lower levels during employing security or protection into the design against various threats such as IP piracy, counterfeiting, cloning and false claim of ownership.
- Since, these IP cores are extensively used in modern electronics devices such as digital camera, laptop, and tablet etc., therefore their protection against false ownership is very crucial.
- Different abstraction levels of the design flow can be leveraged to formulate a protection mechanism.

❑ Sometimes employing protection at higher abstraction levels is more advantageous than lower levels because of the following:

(i) Post- synthesis, security constraints get distributed throughout the design at lower levels.

(ii) Representation of complex designs at higher level offers lesser complexity than lower levels.

(iii) Higher abstraction level allows possibility of design trade-off and optimization of design objectives during embedding watermark.

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- The most common approach used for protecting IP cores against IP piracy, counterfeiting and false claim of ownership is watermarking.
- However, combining two consecutive abstraction levels for implanting watermarking constraints renders the watermark highly robust, thus provides enhanced design protection.
- (Sengupta et al., 2019, Roy and Sengupta, 2019) employed watermark at two consecutive abstraction levels viz. ESL and RTL to protect the DSP IP cores against the aforesaid threats. Such watermark can be referred as multi-level watermark.
- An overview of the possible attacks on IP cores and its protection using multi-level watermarking is shown here:

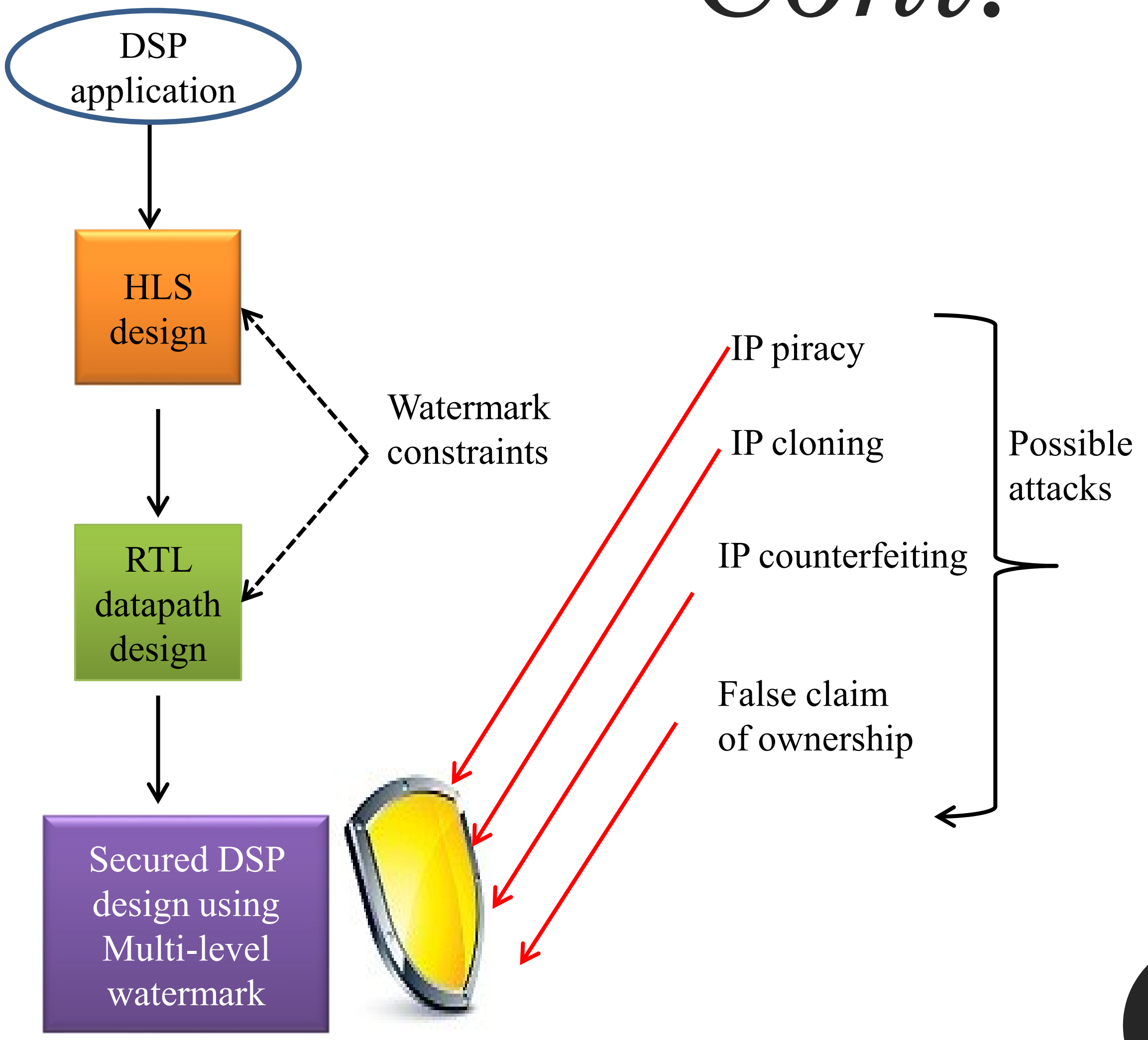

Fig. *Shielding of DSP IP cores against possible attacks (Sengupta et al., 2019)*

Discussion on Selected Approaches

- Approach (Roy and Sengupta, 2018) proposed symmetrical protection for IP cores only at ESL using both seller watermark and buyer fingerprint simultaneously.
- Further, approaches (Koushanfar et al., 2005; Sengupta and Bhadauria, 2016; Sengupta and Roy, 2017) implanted watermark during register allocation phase of high-level synthesis.
- Such watermarks are referred as single-phase watermark as it is implanted in a single phase of an abstraction level.
- (Koushanfar et al., 2005) implanted a dynamic watermark (the watermarking constraints are embedded as pre-processing step before performing the ESL synthesis) by adding some design and timing constraints during ESL (behavioural) synthesis.

❑ (Koushanfar et al., 2005) embedded watermark considering the following objectives:

- (i) embedded watermark should not affect correctness of the functionality
- (ii) it should minimal design overhead
- (iii) it should be transparent in order to be used for other designs and for any CAD tool

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- (iv) the vendor should be capable to prove his/her ownership using watermark
- (v) it should be extremely hard to locate (perceptually invisible) for an attacker
- (vi) it should be resilient against the removal through any mechanism
- (vii) it should be distributive in nature so that not only the entire design but also its various portions could be protected
- (viii) it should be easily detectable at low cost.
- Furthermore, (Sengupta and Roy, 2018; Sengupta et al., 2018) embedded watermark at three different phases (scheduling, hardware allocation, register allocation) of ESL (during high level synthesis). Such watermark is referred as multi-phase watermark.
- However, these watermarks are employed only within a single abstraction level of the design flow.

Process Flow of Watermarking Approach

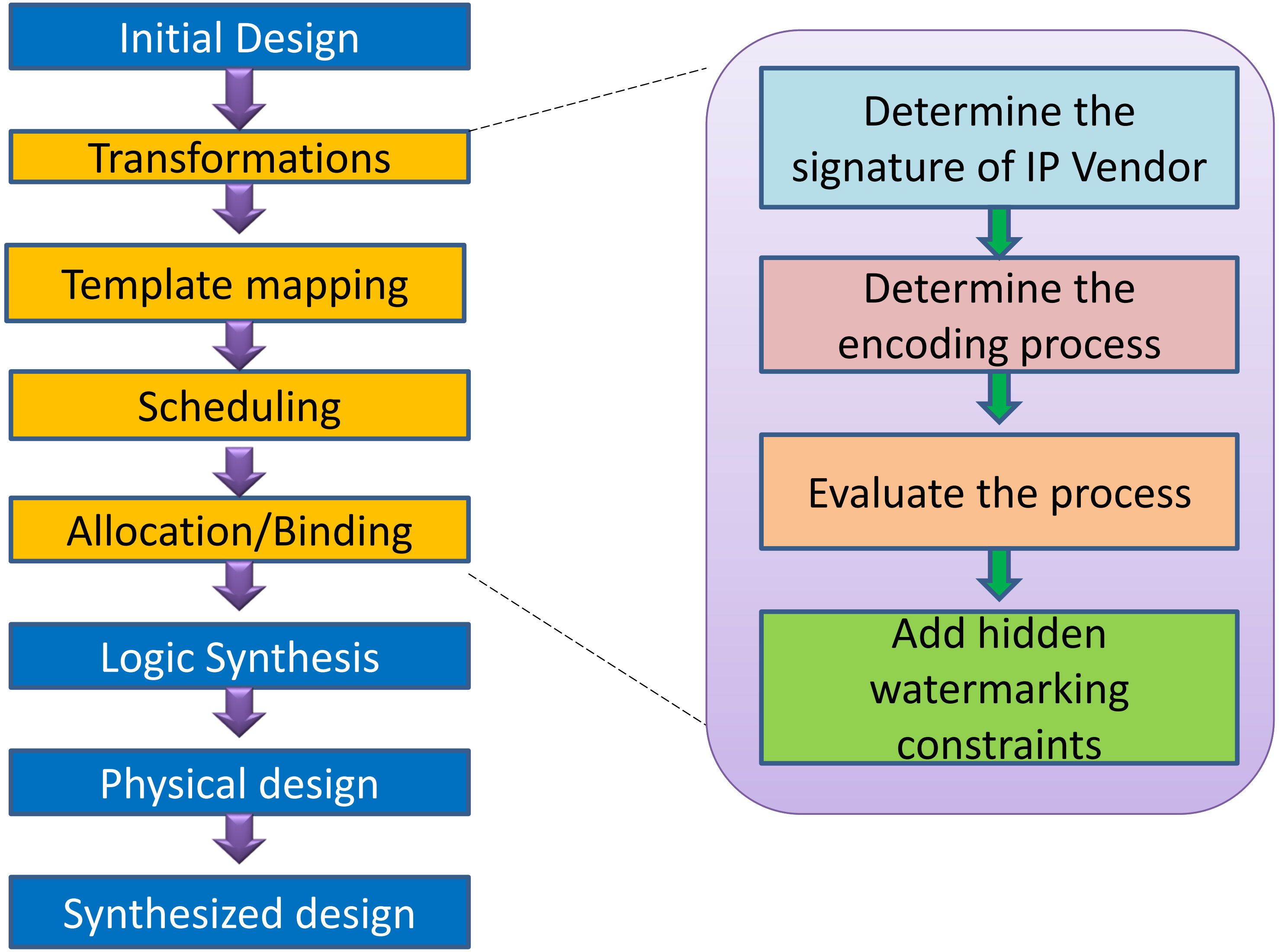

Fig. *Behavioral Synthesis Techniques for Intellectual Property Protection (Koushanfar et al., 2005)*

<u>Encoding process</u>

- ✓ All the nodes in the interval graph are sorted and numbered in the increasing order of the length of their lifetimes.
- ✓ In the increasing order of node numbers, each node is considered for embedding a single bit.
- ✓ After all the nodes are considered, repeat the process from the first node in the order.
- ✓ For each bit in the watermark, the terminal node is chosen such that the terminal node number represents the bit value.
- ✓ To embed 1, the terminal node of the added edge should have an odd number.
- ✓ To embed 0, the terminal node of the added edge should have an even number.

Salient Features and Advantages of Multi-level Watermark

- The multi-level watermark exploits both ESL and RTL of design flow to implant the signature constraints into the IP core design.

❑ This nature of the watermark provides the following salient features and advantages:

- It leverages RTL along with ESL for implanting signature constraints distributivity throughout the design.
- The signature encoding of multi-level watermark is distributive and exclusive by nature.
- Because of covert signature embedding process, higher security and robustness can be achieved.
- The implanted multi-level watermark results into marginal area overhead even with at maximum possible signature strength.

Embedding Signature as Secret-mark

- The overall process of embedding signature as secret-mark is described in following sub-sections.

1) Problem Formulation

- A multi-level watermark protected IP core is designed using an optimal resource configuration.
- A control data flow graph representing a DSP application is given as an input.
- The resource configuration used by the design can be represented using the set X as follows:
  - $X=\{N_1(R_1), N_2(R_2),\ldots, N_m(R_m)\}$
- Where, N_1, N_2 and N_m denote the number of resources of type R_1, R_2 and R_m respectively. Here, m denotes the total number of types of resources.

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2) Details of Multi-level Watermark

a) Overview of the multi-level watermark

- The multi-level watermark is a set of signature constraints that are implanted at two consecutive abstraction levels viz. ESL and RTL.
- The implantation of signature constraints at ESL is performed during **register sharing phase**.
- On the contrary, for implanting signature constraints at RTL, multiplexers (muxes) and de -multiplexers (demuxes) are **fragmented into their next hierarchical size**
- (e.g. one 8:1 mux is fragmented into two 4:1 muxes and one 2: 1 mux, while a one 4:1 mux is fragmented into three 2:1 muxes).

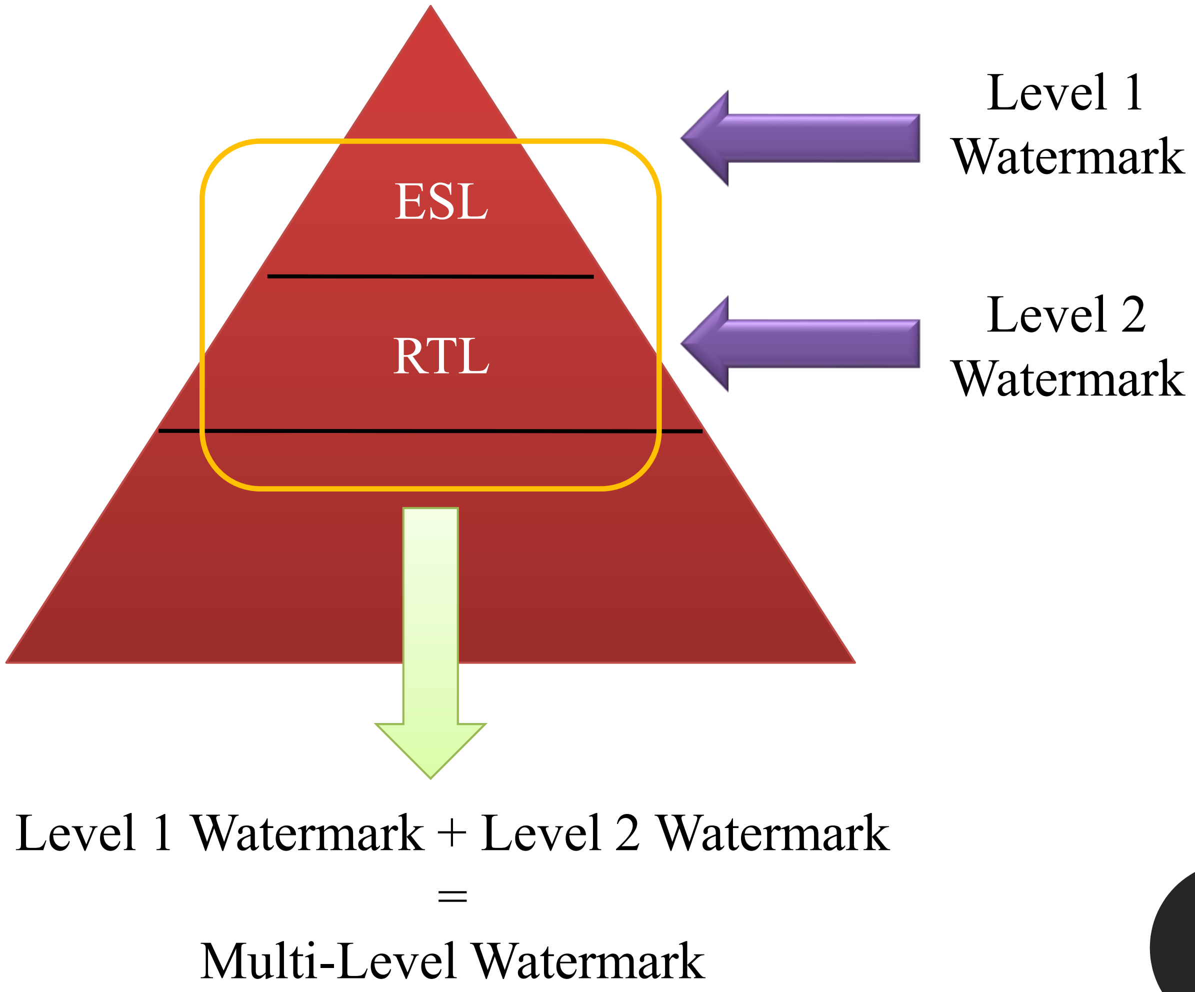

Fig. *Multi- Level Watermarking for DSP cores*

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(b) Signature Encoding

- Three different variables of multi-level watermark signature are 'ω', 'θ' and 'φ'.
- The signature variable 'ω' is implanted into the design during register sharing phase of ESL.
- Further, the signature variables 'θ' and 'φ' are implanted at RTL.

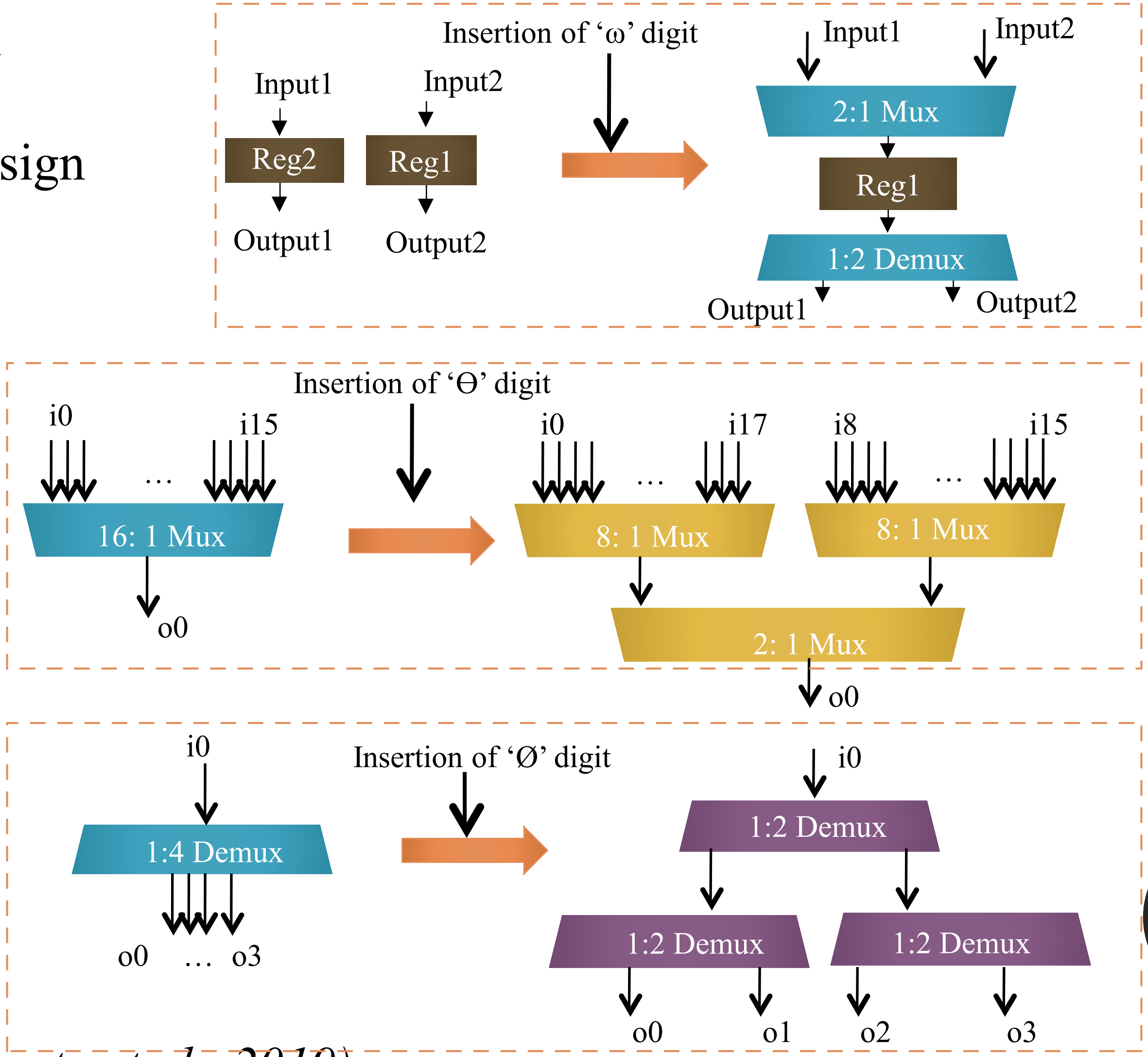

Fig. *Demonstration of watermarking encoding rule (Sengupta et al., 2019)*

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(c) Signature Embedding Process

- ❑ The pre-processing steps in order to implant signature constraints at ESL.

1. Identify the number of registers (in scheduled CDFG) allocated in different control steps.
2. Divide the total number of such registers (identified in step (1)) into group of two or three registers.

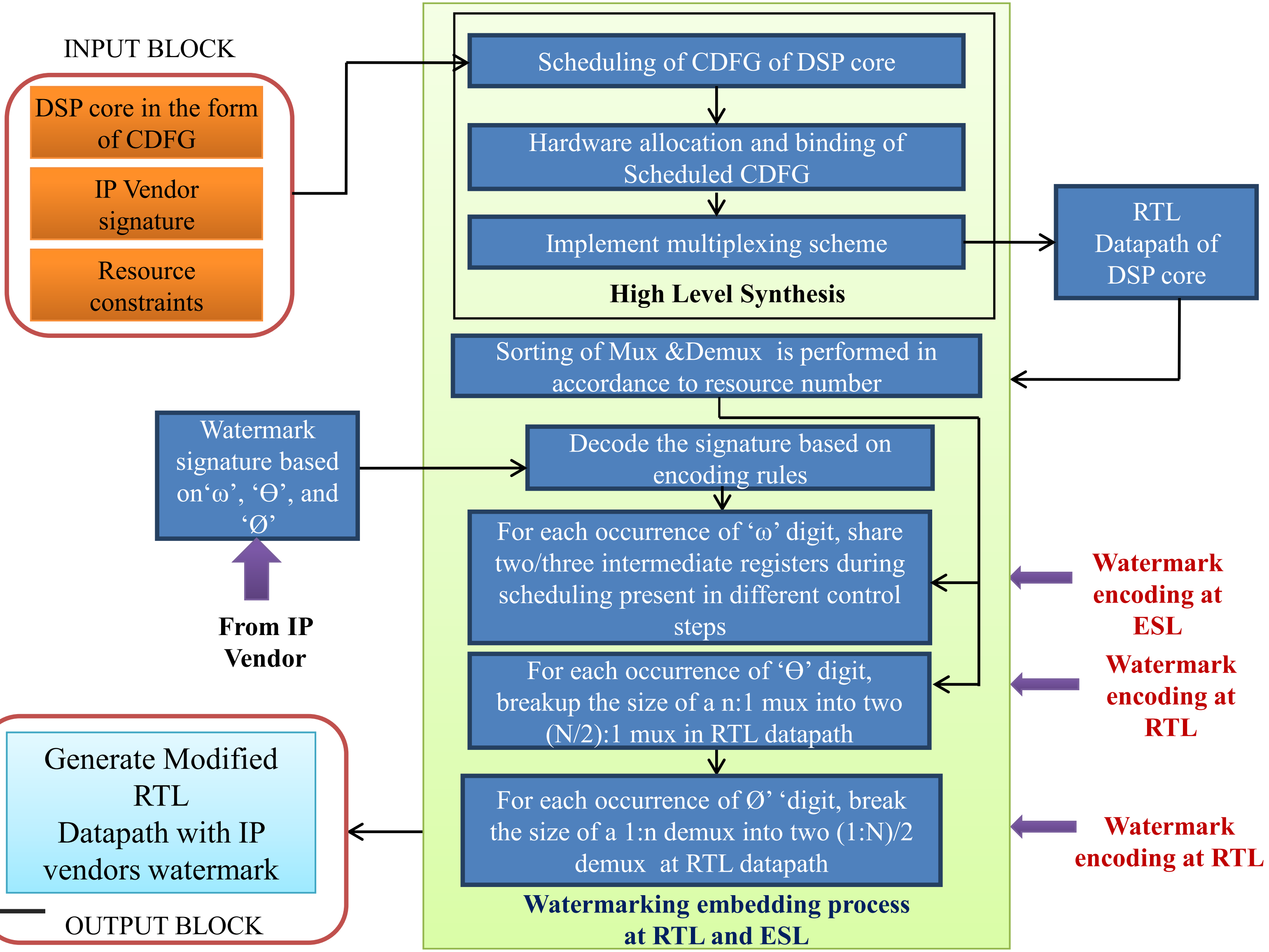

Fig. *Multi-level watermark embedding process at RTL and ESL for DSP cores- (Sengupta et al., 2019)*

Design Process of a Multi-level Watermarked IP Core

- The process of employing multi-level watermark in an IP core is demonstrated using Finite Impulse Response (FIR) filter.
- In order to embed multi-level watermark, a CDFG representing FIR filter is first scheduled using the resource configuration obtained through PSO-DSE as shown below:

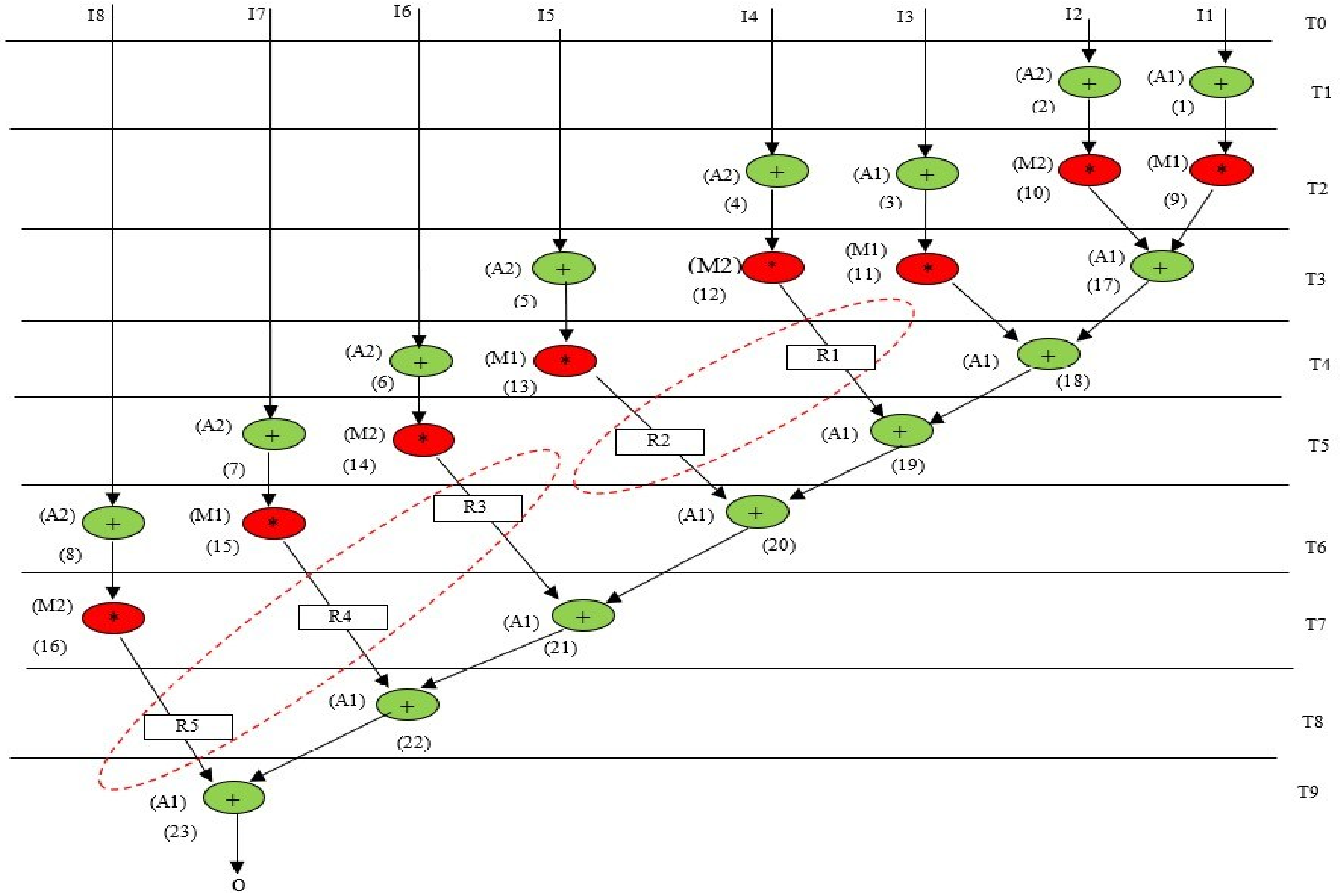

Fig. *Scheduled DFG of FIR filter (Sengupta et al., 2019)*

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- The steps of the multi-level watermark embedding process for the FIR filter are described below:

1. The first step is to identify the total number of the registers allocated in different control steps: (R1, R2, R3, R4 and R5) that have been allocated in different control steps (T4, T5, T6, T7 and T8 respectively).
2. These five registers are distributed into two groups each composed of two/three registers. One group contains R1 and R2 and another group contains R3, R4 and R5.
3. Assuming a vendor selected signature as “θθθθθθθφφφφωω”, Then we perform the fragmentation of muxes and de-muxes.
4. The process of fragmenting in this approach is covert way of inserting vendor signature into the design.
5. This covert way of inserting signature thwarts an attacker from identifying the watermark, thus making it difficult to launch tampering or removal attack.

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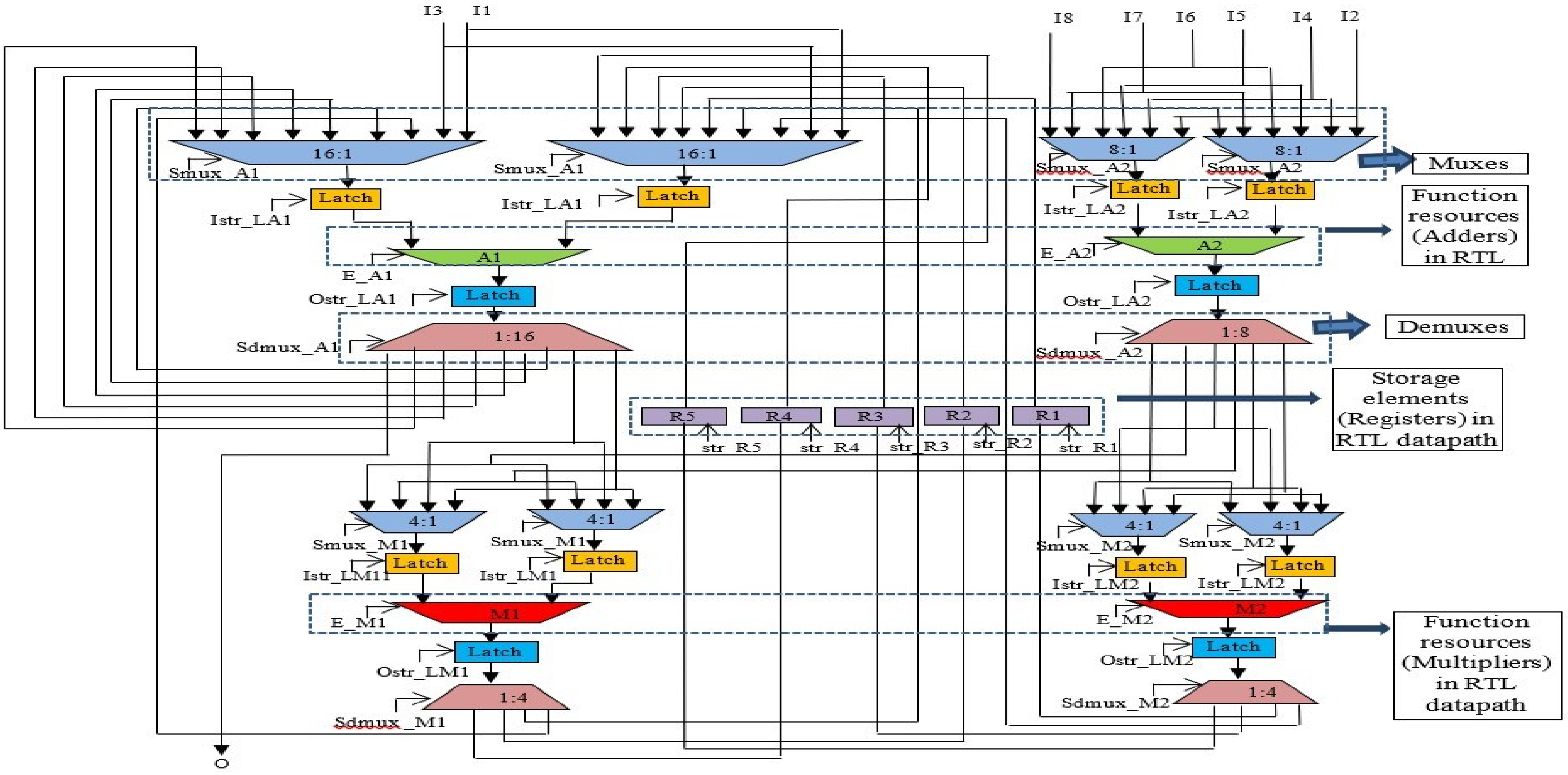

Fig. *FIR filter RTL datapath before watermark (An un-protected RTL datapath of a FIR filter)*

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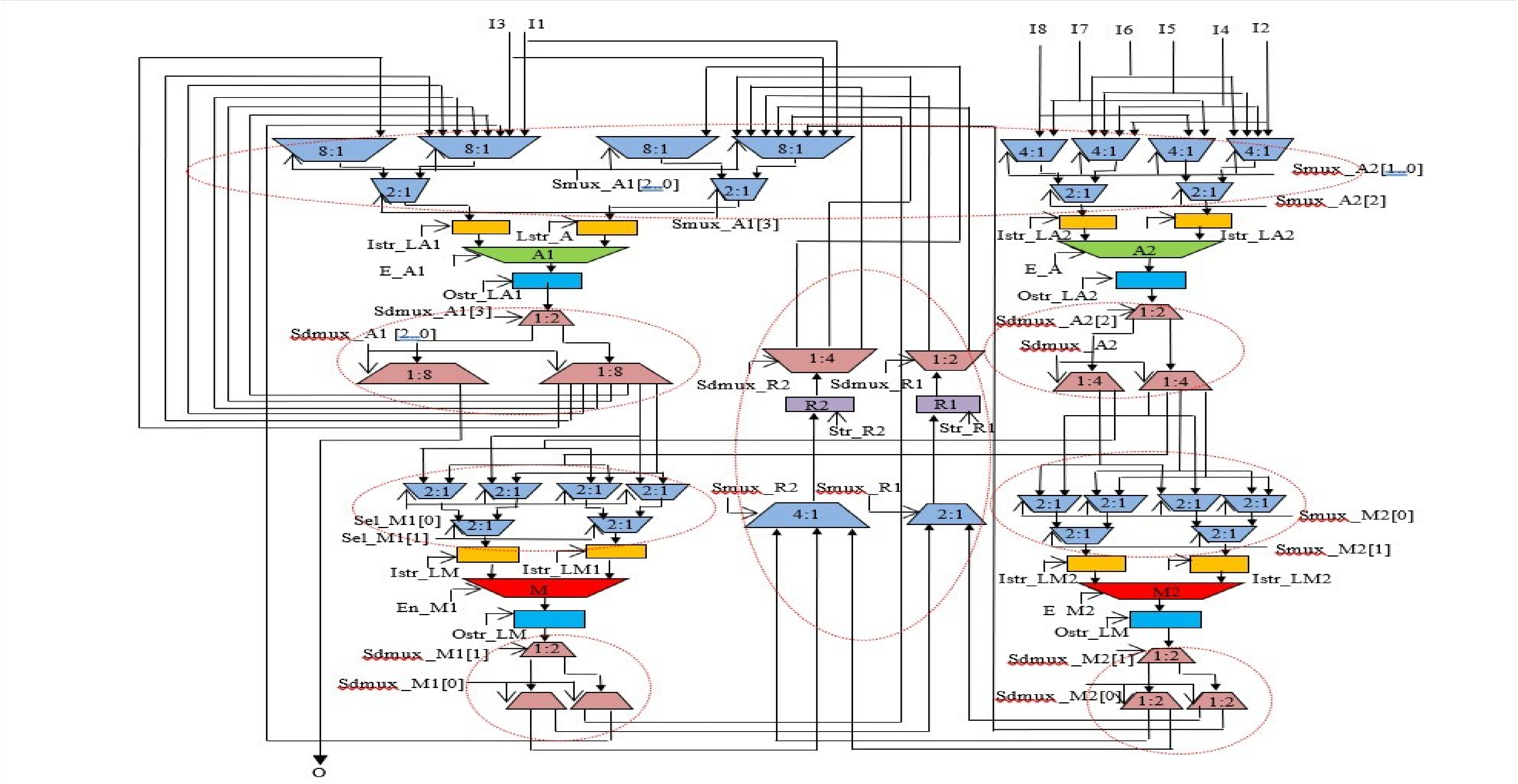
- The impact on the RTL datapath due to watermark has been indicated in dotted red colour ovals as shown below.



Signature Detection in a multi-level watermarked IP core

- The signature detection process is performed through following two sub-steps:

a) Design Inspection:

The RTL design of the IP core is inspected to extract the structural properties of the datapath circuit.

***b) Signature verification*:**

The signature of the vendor is decoded into its respective watermarking constraints and the presence of the constraints is verified in the extracted structural properties of the datapath circuit.

Analysis on Case Studies

- The multi-level watermarked FIR filter has been analysed in terms of security and design cost compared to the baseline design.

1. Security and Design Cost Analysis

- The security of the multi-level watermarked FIR filter is assessed based on the tamper-tolerance ability of the design.
- In order to tamper, the attacker needs to know the following:
- (i) Vendor’s signature size.
- (ii) Number of variables.
- (iii) The encoding of each variable should be known to the attacker.

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- Attacker starts guessing a signature and detecting its presence in the RTL datapath repeatedly until the original signature is not obtained.
- Therefore, the attacker is forced to perform the brute-force approach to find the original signature as shown here.

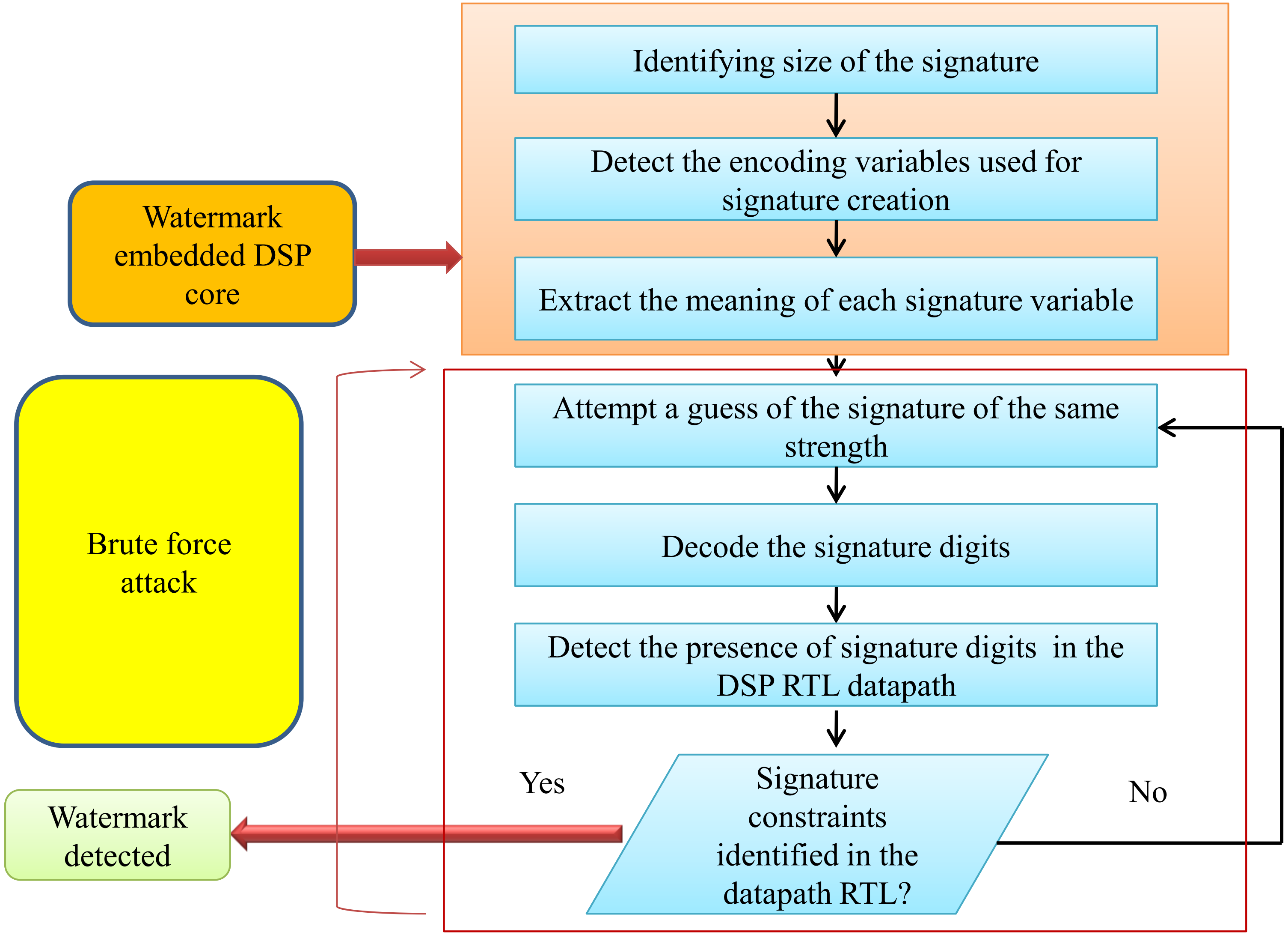

Fig. *Signature detection in multi-phase watermarking using brute-force*

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- The tamper tolerance ability of a multi-watermarked design can be evaluated by calculating the number of all possible signatures using following equation:

$$T_t = k^s \qquad (1)$$

- 'k' denotes the number of encoding variables and 's' denotes the signature size.

Ability of tamper tolerance of multi-level watermarking for different signature sizes				
Signature size	**3**	**6**	**8**	**14**
Signature combination	θθφ	θθθθφφ	θθθθφφωω	θθθθθθθθφφφφωω
Tamper tolerance	27	729	6561	4782972

Design Cost Overhead in Terms of LE

- Furthermore, embedding multi-level watermark into the design may result into overhead hence could affect the design cost.
- The overhead due to embedding the least and largest signature sizes are analysed in terms of logic elements (LE) with respect to a non-watermarked design.

Device utilization summary of 8th order fir filter with no secret signature					
Resource type	**Total Logic Elements**	**Total Combinational Function**	**Dedicated Logic Registers**	**Total Registers**	**Total Pins**
Resource usage	1370/68416 (2%)	1369/68416 (2%)	65/68416 (<1%)	52	145/622 (23%)

Design Cost Overhead in Terms of LE

Summary of device utilization of FIR filter in case of both least and largest possible size signature and comparison of LE overhead in contrast to baseline FIR filter

Signature & resource details	Details of maximum signature	Details of Minimum signature
Signature size	14	3
Signature digits	θθθθθθθθφφφφωω	θθφ
Total pin	145/622 (23%)	145/622 (23%)
Dedicated logic registers	65/68416 (<1%)	65/68416 (<1%)
Total register	52	52
Total combinational function	1458/68416 (2%)	1417/68416 (2%)
Total logic elements	1461/68416 (2%)	1418/68416 (2%)
Overhead (%)	6%	3%

Conclusion

- ✓ An approach of deploying protection (using watermark) at two consecutive abstraction levels (ESL and RTL) has been presented.
- ✓ Because of presence of the watermark at two abstraction levels, detection and tampering of the signature by an adversary become highly complex.
- ✓ However, the multi-level watermark can easily be extracted during forensic detective control in case of false claim of ownership and IP piracy/counterfeiting/cloning.
- ✓ Higher strength of watermark is achieved at nominal design overhead.

References

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Thank You